



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

REINFORCEMENT LEARNING PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A AI/ML

TOTAL: 10 QUESTIONS

Q1 What is Reinforcement Learning and what AI/ML use case is it designed for?

Q6 How do you evaluate a model's performance in Reinforcement Learning?

Q2 What are the core abstractions in Reinforcement Learning?

Q7 How do you deploy a model trained with Reinforcement Learning?

Q3 How does Reinforcement Learning handle training or inference?

Q8 How does Reinforcement Learning handle distributed or multi-GPU training?

Q4 How does Reinforcement Learning manage data preprocessing?

Q9 How do you track experiments and versions in Reinforcement Learning?

Q5 How does Reinforcement Learning support GPU/TPU acceleration?

Q10 What are common pitfalls when using Reinforcement Learning in production?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Reinforcement Learning is a framework or service

Mitigation

Reinforcement Learning is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set and

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 Reinforcement Learning organizes work around abstractions

Abstractions

Reinforcement Learning organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A7 Export the model to a portable format

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, Reinforcement Learning defines a model

Hardening

For training, Reinforcement Learning defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 Reinforcement Learning provides data-parallel or

Governance

Reinforcement Learning provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 Reinforcement Learning provides data loaders, tokenizers, or

Transformers

Reinforcement Learning provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights & Biases, Comet)

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside Reinforcement Learning to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 Reinforcement Learning detects available accelerators and

Optimization

Reinforcement Learning detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew (different preprocessing)

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.